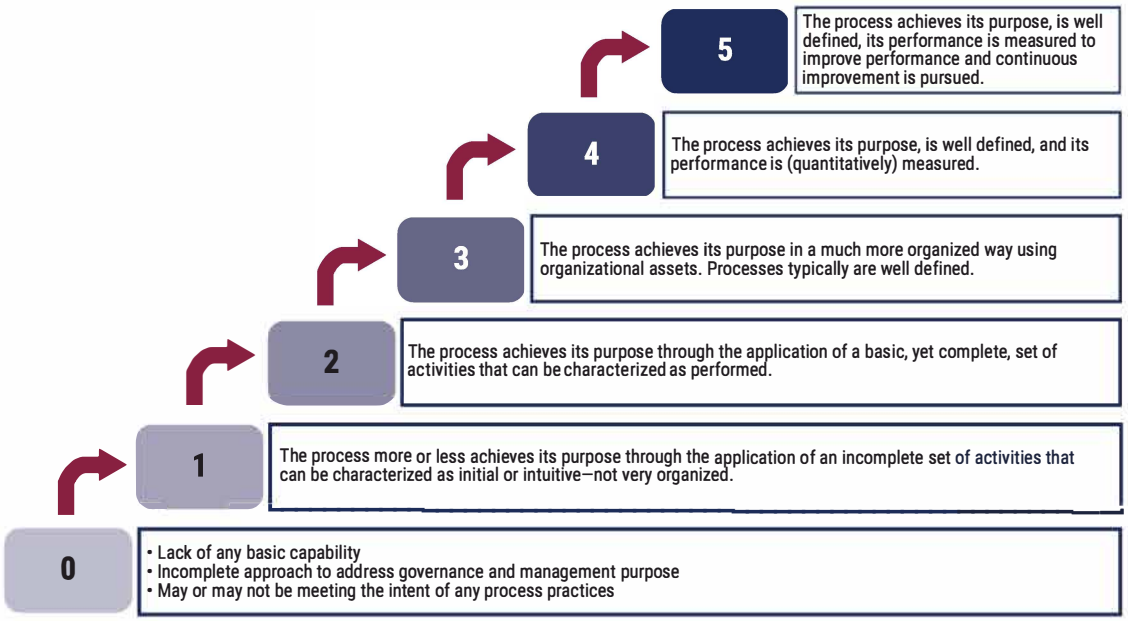
Figure 6.2—Capability Levels for Processes



The COBIT core model assigns capability levels to all process activities, enabling clear definition of the processes and required activities for achieving the different capability levels. See *COBIT® 2019 Framework: Governance and Management Objectives* for more detail.

6.4.2 Rating Process Activities

A capability level can be achieved to varying degrees, which can be expressed by a set of ratings. The range of available ratings depends on the context in which the performance assessment is made:

- Some formal methods leading to independent certification use a binary pass/fail set of ratings.
- Less formal methods (often used in performance-improvement contexts) work better with a larger range of ratings, such as the following set:
  - *Fully*—The capability level is achieved for more than 85 percent. (This remains a judgment call, but it can be substantiated by the examination or assessment of the components of the enabler, such as process activities, process goals or organizational structure good practices.)
  - *Largely*—The capability level is achieved between 50 percent and 85 percent.
  - *Partially*—The capability level is achieved between 15 percent and 50 percent.
  - *Not*—The capability level is achieved less than 15 percent.

6.4.3 Focus Area Maturity Levels

Sometimes a higher level is required for expressing performance without the granularity applicable to individual process capability ratings. Maturity levels can be used for that purpose. COBIT® 2019 defines maturity levels as a performance measure at the focus area level, as shown in **figure 6.3**.

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Domain: Align, Plan and Organize Management Objective: APO13 — Managed Security	Focus Area: COBIT Core Model

Description

Define, operate and monitor an information security management system.

Purpose

Keep the impact and occurrence of information security incidents within the enterprise's risk appetite levels.

The management objective supports the achievement of a set of primary enterprise and alignment goals:

Enterprise Goals	Alignment Goals
• EG02 Managed business risk • EG06 Business service continuity and availability	AG07 Security of information, processing infrastructure and applications, and privacy
Example Metrics for Enterprise Goals	**Example Metrics for Alignment Goals**
EG02 a. Percent of critical business objectives and services covered by risk assessment b. Ratio of significant incidents that were not identified in risk assessments vs. total incidents c. Frequency of updating risk profile	AG07 a. Number of confidentiality incidents causing financial loss, business disruption or public embarrassment b. Number of availability incidents causing financial loss, business disruption or public embarrassment c. Number of integrity incidents causing financial loss, business disruption or public embarrassment
EG06 a. Number of customer service or business process interruptions causing significant incidents b. Business cost of incidents c. Number of business processing hours lost due to unplanned service interruptions d. Percent of complaints as a function of committed service availability targets	

A. Component: Process

Management Practice	Example Metrics
APO13.01 Establish and maintain an information security management system (ISMS). Establish and maintain an information security management system (ISMS) that provides a standard, formal and continuous approach to information security management, enabling secure technology and business processes that are aligned with business requirements.	a. Level of stakeholder satisfaction with the security plan throughout the enterprise

Activities	Capability Level
1. Define the scope and boundaries of the information security management system (ISMS) in terms of the characteristics of the enterprise, the organization, its location, assets and technology. Include details of, and justification for, any exclusions from the scope.	2
2. Define an ISMS in accordance with enterprise policy and the context in which the enterprise operates.	
3. Align the ISMS with the overall enterprise approach to the management of security.	
4. Obtain management authorization to implement and operate or change the ISMS.	
5. Prepare and maintain a statement of applicability that describes the scope of the ISMS.	
6. Define and communicate Information security management roles and responsibilities.	
7. Communicate the ISMS approach.	

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A. Component: Process *(cont.)*	
Related Guidance (Standards, Frameworks, Compliance Requirements)	**Detailed Reference**
HITRUST CSF version 9, September 2017	0.01 Information Security Management program
ISO/IEC 20000-1:2011(E)	6.6 Information security management
ITIL V3, 2011	Service Design, 4.7 Information Security Management
National Institute of Standards and Technology Special Publication 800-37, Revision 2 (Draft), May 2018	3.3 Selection (Task 1); 3.4 Implementation (Task 1)
National Institute of Standards and Technology Special Publication 800-53, Revision 5 (Draft), August 2017	3.17 Risk assessment (RA-2)
Management Practice	**Example Metrics**
APO13.02 Define and manage an information security and privacy risk treatment plan. Maintain an information security plan that describes how information security risk is to be managed and aligned with enterprise strategy and enterprise architecture. Ensure that recommendations for implementing security improvements are based on approved business cases, implemented as an integral part of services and solutions development, and operated as an integral part of business operation.	a. Percentage of successful security risk scenario simulations b. Number of employees who have successfully completed information security awareness training

Activities	Capability Level
1. Formulate and maintain an information security risk treatment plan aligned with strategic objectives and the enterprise architecture. Ensure that the plan identifies the appropriate and optimal management practices and security solutions, with associated resources, responsibilities and priorities for managing identified information security risk.	3
2. Maintain as part of the enterprise architecture an inventory of solution components that are in place to manage security-related risk.	
3. Develop proposals to implement the information security risk treatment plan, supported by suitable business cases that include consideration of funding and allocation of roles and responsibilities.	
4. Provide input to the design and development of management practices and solutions selected from the information security risk treatment plan.	
5. Implement information security and privacy training and awareness programs.	
6. Integrate the planning, design, implementation and monitoring of information security and privacy procedures and other controls capable of enabling prompt prevention, detection of security events, and response to security incidents.	
7. Define how to measure the effectiveness of the selected management practices. Specify how these measurements are to be used to assess effectiveness to produce comparable and reproducible results.	4

Related Guidance (Standards, Frameworks, Compliance Requirements)	Detailed Reference
No related guidance for this management practice	
Management Practice	**Example Metrics**
APO13.03 Monitor and review the information security management system (ISMS). Maintain and regularly communicate the need for, and benefits of, continuous improvement in information security. Collect and analyze data about the information security management system (ISMS), and improve its effectiveness. Correct nonconformities to prevent recurrence.	a. Frequency of scheduled security reviews b. Number of findings in regularly scheduled security reviews c. Level of stakeholder satisfaction with the security plan d. Number of security-related incidents caused by failure to adhere to the security plan

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A. Component: Process *(cont.)*	
Activities	**Capability Level**
1. Undertake regular reviews of the effectiveness of the ISMS. Include meeting ISMS policy and objectives and reviewing security and privacy practices.	4
2. Conduct ISMS audits at planned intervals.	
3. Undertake a management review of the ISMS on a regular basis to ensure that the scope remains adequate and improvements in the ISMS process are identified.	
4. Record actions and events that could have an impact on the effectiveness or performance of the ISMS.	
5. Provide input to the maintenance of the security plans to take into account the findings of monitoring and reviewing activities.	5

Related Guidance (Standards, Frameworks, Compliance Requirements)	Detailed Reference
National Institute of Standards and Technology Special Publication 800-37, Revision 2 (Draft), May 2018	3.3 Selection (Task 3)

B. Component: Organizational Structures

Key Management Practice	Chief Information Officer	Chief Technology Officer	Enterprise Risk Committee	Chief Information Security Officer	Business Process Owners	Project Management Office	Head Architect	Head Development	Head IT Operations	Head IT Administration	Service Manager	Information Security Manager	Business Continuity Manager	Privacy Officer
APO13.01 Establish and maintain an information security management system (ISMS).	R		R	A						R		R		
APO13.02 Define and manage an information security and privacy risk treatment plan.	R		R	A						R		R		R
APO13.03 Monitor and review the information security management system (ISMS).	R	R		A	R	R	R	R	R	R	R	R	R	R

Related Guidance (Standards, Frameworks, Compliance Requirements)	Detailed Reference
ISF, The Standard of Good Practice for Information Security 2016	SG1.2 Security Direction
ISO/IEC 27002:2013/Cor.2:2015(E)	6.1 Internal organization

C. Component: Information Flows and Items (see also Section 3.6)

Management Practice	Inputs		Outputs	
	From	**Description**	**Description**	**To**
APO13.01 Establish and maintain an information security management system (ISMS).	Outside COBIT	Enterprise security approach	ISMS scope statement	APO01.05; DSS06.03
			ISMS policy	Internal
APO13.02 Define and manage an information security risk treatment plan.	APO02.04	Gaps and changes required to realize target capability	Information security risk treatment plan	All APO; All BAI; All DSS; All MEA; ALL EDM
	APO03.02	Baseline domain descriptions and architecture definition	Information security business cases	APO05.02
	APO12.05	Project proposals for reducing risk		

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C. Component: Information Flows and Items (see also Section 3.6) *(cont.)*

Management Practice	Inputs		Outputs	
APO13.03 Monitor and review the information security management system (ISMS).	**From**	**Description**	**Description**	**To**
	DSS02.02	Classified and prioritized incidents and service requests	Recommendations for improving the information security management system (ISMS)	Internal
			Information security management system (ISMS) audit reports	MEA02.01

Related Guidance (Standards, Frameworks, Compliance Requirements)	Detailed Reference
National Institute of Standards and Technology Special Publication 800-37, Revision 2, September 2017	3.3 Selection (Tasks 1, 3): Inputs and Outputs; 3.4 Implementation (Task 1): Inputs and Outputs

D. Component: People, Skills and Competencies

Skill	Related Guidance (Standards, Frameworks, Compliance Requirements)	Detailed Reference
Information security	Skills Framework for the Information Age V6, 2015	SCTY
Information security strategy development	e-Competence Framework (e-CF)—A common European Framework for ICT Professionals in all industry sectors - Part 1: Framework, 2016	D. Enable—D.1. Information Security Strategy Development

E. Component: Policies and Procedures

Relevant Policy	Policy Description	Related Guidance	Detailed Reference
Information security and privacy policy	Sets behavioral guidelines to protect corporate information, systems and infrastructure. Given that business requirements regarding security and storage are more dynamic than I&T risk management and privacy, their governance should be handled separately from that of I&T risk and privacy. For operational efficiency, synchronize information security policy with I&T risk and privacy policy.	(1) ISO/IEC 27001:2013/Cor.2:2015(E); (2) ISO/IEC 27002:2013/Cor.2:2015(E); (3) National Institute of Standards and Technology Special Publication 800-53, Revision 5 (Draft), August 2017; (4) HITRUST CSF version 9, September 2017; (5) ISF, The Standard of Good Practice for Information Security 2016	(1) 5.2 Policy; (2) 5. Information security policies; (3) 3.2 Awareness and training (AT-1); (4) 04.01 Information Security Policy; (5) SM1.1 Information Security Policy

F. Component: Culture, Ethics and Behavior

Key Culture Elements	Related Guidance	Detailed Reference
Establish a culture of security and privacy awareness that positively influences desirable behavior and actual implementation of security and privacy policy in daily practice. Provide sufficient security and privacy guidance, indicate security and privacy champions (including C-level executives, leaders in HR, and security and/or privacy professionals) and proactively support and communicate security and privacy programs, innovations and challenges.	(1) ISO/IEC 27001:2013/Cor.2:2015(E); (2) Creating a Culture of Security, ISACA, 2011	1) 7.3 Awareness; (2) Framework to achieve an intentional security aware culture (all chapters)

G. Component: Services, Infrastructure and Applications

- Configuration management tools
- Security and privacy awareness services
- Third-party security assessment services

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Domain: Deliver, Service and Support Management Objective: DSS05 - Managed Security Services	Focus Area: COBIT Core Model

Description

Protect enterprise information to maintain the level of information security risk acceptable to the enterprise in accordance with the security policy. Establish and maintain information security roles and access privileges. Perform security monitoring.

Purpose

Minimize the business impact of operational information security vulnerabilities and incidents.

The management objective supports the achievement of a set of primary enterprise and alignment goals:

Enterprise Goals	Alignment Goals
• EG02 Managed business risk • EG06 Business service continuity and availability	• AG02 Managed I&T-related risk • AG07 Security of information, processing infrastructure and applications, and privacy
Example Metrics for Enterprise Goals	**Example Metrics for Alignment Goals**
EG02 a. Percent of critical business objectives and services covered by risk assessment b. Ratio of significant incidents that were not identified in risk assessments vs. total incidents c. Frequency of updating risk profile	AG02 a. Frequency of updating risk profile b. Percent of enterprise risk assessments including I&T-related risk c. Number of significant I&T-related incidents that were not identified in a risk assessment
EG06 a. Number of customer service or business process interruptions causing significant incidents b. Business cost of incidents c. Number of business processing hours lost due to unplanned service interruptions d. Percent of complaints as a function of committed service availability targets	AG07 a. Number of confidentiality incidents causing financial loss, business disruption or public embarrassment b. Number of availability incidents causing financial loss, business disruption or public embarrassment c. Number of integrity incidents causing financial loss, business disruption or public embarrassment

A. Component: Process

Management Practice	Example Metrics
DSS05.01 Protect against malicious software. Implement and maintain preventive, detective and corrective measures (especially up-to-date security patches and virus control) across the enterprise to protect information systems and technology from malicious software (e.g., ransomware, malware, viruses, worms, spyware, spam).	a. Number of successful malicious software attacks b. Percent of employees failing tests on malicious attacks (e.g., test of phishing email)

Activities	Capability Level
1. Install and activate malicious software protection tools on all processing facilities, with malicious software definition files that are updated as required (automatically or semi-automatically).	2
2. Filter incoming traffic, such as email and downloads, to protect against unsolicited information (e.g., spyware, phishing emails).	
3. Communicate malicious software awareness and enforce prevention procedures and responsibilities. Conduct periodic training about malware in email and Internet usage. Train users to not open, but report, suspicious emails and to not install shared or unapproved software.	3
4. Distribute all protection software centrally (version and patch-level) using centralized configuration and IT change management.	
5. Regularly review and evaluate information on new potential threats (e.g., reviewing vendors' products and services security advisories).	4

Related Guidance (Standards, Frameworks, Compliance Requirements)	Detailed Reference
CMMI Cybermaturity Platform, 2018	DP.DC Detect Malicious Code; RI.VT Vulnerability and Threat Identification
HITRUST CSF version 9, September 2017	09.04 Protection Against Malicious & Mobile Code
SF, The Standard of Good Practice for Information Security 2016	TS1 Security Solutions
SO/IEC 27002:2013/Cor.2:2015(E)	12.2 Protection against malware
The CIS Critical Security Controls for Effective Cyber Defense Version 6.1, August 2016	CSC 4: Continuous Vulnerability Assessment and Remediation; CSC 8: Malware Defenses

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A. Component: Process *(cont.)*	
Management Practice	**Example Metrics**
DSS05.02 Manage network and connectivity security. Use security measures and related management procedures to protect information over all methods of connectivity.	a. Number of firewall breaches b. Number of vulnerabilities discovered c. Percent of time network and systems not available due to security incident

Activities	Capability Level
1. Allow only authorized devices to have access to corporate information and the enterprise network. Configure these devices to force password entry.	2
2. Implement network filtering mechanisms, such as firewalls and intrusion detection software. Enforce appropriate policies to control inbound and outbound traffic.	
3. Apply approved security protocols to network connectivity.	
4. Configure network equipment in a secure manner.	
5. Encrypt information in transit according to its classification.	3
6. Based on risk assessments and business requirements, establish and maintain a policy for security of connectivity.	
7. Establish trusted mechanisms to support the secure transmission and receipt of information.	
8. Carry out periodic penetration testing to determine adequacy of network protection.	4
9. Carry out periodic testing of system security to determine adequacy of system protection.	

Related Guidance (Standards, Frameworks, Compliance Requirements)	Detailed Reference
CMMI Cybermaturity Platform, 2018	AC.MI Manage Network Integrity & Segregation; CM.MN Monitor Networks; AC.CP Manage Communication Protections
HITRUST CSF version 9, September 2017	01.04 Network Access Control
ISF, The Standard of Good Practice for Information Security 2016	PA2.3 Mobile Device Connectivity; NC1.1 Network Device Configuration
ISO/IEC 27002:2013/Cor.2:2015(E)	13.1 Network security management
National Institute of Standards and Technology Special Publication 800-53, Revision 5 (Draft), August 2017	3.20 System and information integrity (SI-8)
The CIS Critical Security Controls for Effective Cyber Defense Version 6.1, August 2016	CSC 9: Limitation and Control of Network Ports, Protocols, and Services; CSC 11: Secure Configurations for Network Devices such as Firewalls, Routers, and Switches

Management Practice	Example Metrics
DSS05.03 Manage endpoint security. Ensure that endpoints (e.g., laptop, desktop, server, and other mobile and network devices or software) are secured at a level that is equal to or greater than the defined security requirements for the information processed, stored or transmitted.	a. Number of incidents involving endpoint devices b. Number of unauthorized devices detected on the network or in the end-user environment c. Percent of individuals receiving awareness training relating to use of endpoint devices

Activities	Capability Level
1. Configure operating systems in a secure manner.	2
2. Implement device lockdown mechanisms.	
3. Manage remote access and control (e.g., mobile devices, teleworking).	
4. Manage network configuration in a secure manner.	
5. Implement network traffic filtering on endpoint devices.	
6. Protect system integrity.	
7. Provide physical protection of endpoint devices.	
8. Dispose of endpoint devices securely.	
9. Manage malicious access through email and web browsers. For example, block certain websites and deactivate click-through on links for smartphones.	
10. Encrypt information in storage according to its classification.	3

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A. Component: Process *(cont.)*	
Related Guidance (Standards, Frameworks, Compliance Requirements)	**Detailed Reference**
CMMI Cybermaturity Platform, 2018	IP.MM Apply Mobile Device Management; TP.MP Apply Media Protection; DP.DP Detect Mobile Code and Browser Protection
ISF, The Standard of Good Practice for Information Security 2016	PM1.3 Remote Working; PA2.1 Mobile Device Configuration; PA2.4 Employee-owned Devices; PA2.5 Portable Storage Devices; NC1.6 Remote Maintenance
National Institute of Standards and Technology Special Publication 800-53, Revision 5 (Draft), August 2017	3.4 Assessment, authorization and monitoring (CA-8, CA-9); 3.19 System and communications protection (SC-10)
The CIS Critical Security Controls for Effective Cyber Defense Version 6.1, August 2016	CSC 3: Secure Configurations for Hardware and Software on Mobile Devices, Laptops, Workstations, and Servers; CSC 7: Email and Web Browser Protections
Management Practice	**Example Metrics**
DSS05.04 Manage user identity and logical access. Ensure that all users have information access rights in accordance with business requirements. Coordinate with business units that manage their own access rights within business processes.	a. Average time between change and update of accounts b. Number of accounts (vs. number of authorized users/staff) c. Number of incidents relating to unauthorized access to information

Activities	Capability Level
1. Maintain user access rights in accordance with business function, process requirements and security policies. Align the management of identities and access rights to the defined roles and responsibilities, based on least-privilege, need-to-have and need-to-know principles.	2
2. Administer all changes to access rights (creation, modifications and deletions) in a timely manner based only on approved and documented transactions authorized by designated management individuals.	3
3. Segregate, reduce to the minimum number necessary and actively manage privileged user accounts. Ensure monitoring on all activity on these accounts.	
4. Uniquely identify all information processing activities by functional roles. Coordinate with business units to ensure that all roles are consistently defined, including roles that are defined by the business itself within business process applications.	
5. Authenticate all access to information assets based on the individual's role or business rules. Coordinate with business units that manage authentication within applications used in business processes to ensure that authentication controls have been properly administered.	
6. Ensure that all users (internal, external and temporary) and their activity on IT systems (business application, IT infrastructure, system operations, development and maintenance) are uniquely identifiable.	
7. Maintain an audit trail of access to information depending upon its sensitivity and regulatory requirements.	4
8. Perform regular management review of all accounts and related privileges.	

Related Guidance (Standards, Frameworks, Compliance Requirements)	Detailed Reference
HITRUST CSF version 9, September 2017	10.03 Cryptographic Controls
ISF, The Standard of Good Practice for Information Security 2016	PM1.1 Employment Life Cycle; SA1 Access Management
ISO/IEC 27002:2013/Cor.2:2015(E)	7.3 Termination and change of employment; 9. Access control
ITIL V3, 2011	Service Operation, 4.5 Access Management
National Institute of Standards and Technology Special Publication 800-53, Revision 5 (Draft), August 2017	3.1 Access control (AC-11, AC-12); 3.11 Media protection (MP-2, MP-4, MP-7); 3.13 Physical and environmental protection (PE-2, PE-3, PE-6)
The CIS Critical Security Controls for Effective Cyber Defense Version 6.1, August 2016	CSC 1: Inventory of Authorized and Unauthorized Devices; CSC 2: Inventory of Authorized and Unauthorized Software; CSC 5: Controlled Use of Administrative Privileges; CSC 16: Account Monitoring and Control

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A. Component: Process *(cont.)*	
Management Practice	**Example Metrics**
DSS05.05 Manage physical access to I&T assets. Define and implement procedures (including emergency procedures) to grant, limit and revoke access to premises, buildings and areas, according to business need. Access to premises, buildings and areas should be justified, authorized, logged and monitored. This requirement applies to all persons entering the premises, including staff, temporary staff, clients, vendors, visitors or any other third party.	a. Average rating for physical security assessments b. Number of physical information security-related incidents

Activities	Capability Level
1. Log and monitor all entry points to IT sites. Register all visitors, including contractors and vendors, to the site.	2
2. Ensure all personnel display properly approved identification at all times.	
3. Require visitors to be escorted at all times while on-site.	
4. Restrict and monitor access to sensitive IT sites by establishing perimeter restrictions, such as fences, walls and security devices on interior and exterior doors.	
5. Manage requests to allow appropriately authorized access to the computing facilities.	3
6. Ensure that access profiles remain current. Base access to IT sites (server rooms, buildings, areas or zones) on job function and responsibilities.	
7. Conduct regular physical information security awareness training.	

Related Guidance (Standards, Frameworks, Compliance Requirements)	Detailed Reference
CMMI Cybermaturity Platform, 2018	AC.MA Manage Access; ID.DI Determine Impacts
HITRUST CSF version 9, September 2017	01.01 Business Requirement for Access Control; 01.02 Authorized Access to Information Systems; 02.0 Human Resources Security
ISF, The Standard of Good Practice for Information Security 2016	NC1.2 Physical Network Management
ISO/IEC 27002:2013/Cor.2:2015(E)	11. Physical and environmental security

Management Practice	Example Metrics
DSS05.06 Manage sensitive documents and output devices. Establish appropriate physical safeguards, accounting practices and inventory management regarding sensitive I&T assets, such as special forms, negotiable instruments, special-purpose printers or security tokens.	a. Number of stolen output devices b. Percent of sensitive documents and output devices identified in inventory

Activities	Capability Level
1. Establish procedures to govern the receipt, use, removal and disposal of sensitive documents and output devices into, within, and outside of the enterprise.	2
2. Ensure cryptographic controls are in place to protect sensitive electronically stored information.	
3. Assign access privileges to sensitive documents and output devices based on the least-privilege principle, balancing risk and business requirements.	3
4. Establish an inventory of sensitive documents and output devices, and conduct regular reconciliations.	
5. Establish appropriate physical safeguards over sensitive documents.	

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A. Component: Process *(cont.)*	
Related Guidance (Standards, Frameworks, Compliance Requirements)	**Detailed Reference**
CMMI Cybermaturity Platform, 2018	CM.Ph Monitor Physical
HITRUST CSF version 9, September 2017	01.06 Application & Information Access Control; 01.07 Mobile Computing & Teleworking; 08.0 Physical & Environmental Security; 10.03 Cryptographic Controls; 10.04 Security of System Files
ISF, The Standard of Good Practice for Information Security 2016	IR2.3 Business Impact Assessment - Confidentiality Requirements; IR2.4 Business Impact Assessment - Integrity Requirements; IR2.5 Business Impact Assessment - Availability Requirements; IM2.2 Sensitive Physical Information; PA2.2 Enterprise Mobility Man
ISO/IEC 27002:2013/Cor.2:2015(E)	10. Cryptography
National Institute of Standards and Technology Special Publication 800-53, Revision 5 (Draft), August 2017	3.1 Access control (AC-2, AC-3, AC-4, AC-5, AC-6, AC-13, AC-24); 3.7 Identification and authentication (IA-2, IA-10, IA-11)
The CIS Critical Security Controls for Effective Cyber Defense Version 6.1, August 2016	CSC 15: Wireless Access Control
Management Practice	**Example Metrics**
DSS05.07 Manage vulnerabilities and monitor the infrastructure for security-related events. Using a portfolio of tools and technologies (e.g., intrusion detection tools), manage vulnerabilities and monitor the infrastructure for unauthorized access. Ensure that security tools, technologies and detection are integrated with general event monitoring and incident management.	a. Number of vulnerability tests carried out on perimeter devices b. Number of vulnerabilities discovered during testing c. Time taken to remediate any vulnerabilities d. Percent of tickets created in a timely manner when monitoring systems identify potential security incidents

Activities	Capability Level
1. Continually use a portfolio of supported technologies, services and assets (e.g., vulnerability scanners, fuzzers and sniffers, protocol analyzers) to identify information security vulnerabilities.	2
2. Define and communicate risk scenarios, so they can be easily recognized, and the likelihood and impact understood.	
3. Regularly review the event logs for potential incidents.	
4. Ensure that security--related incident tickets are created in a timely manner when monitoring identifies potential incidents.	
5. Log security-related events and retain records for appropriate period.	3

Related Guidance (Standards, Frameworks, Compliance Requirements)	Detailed Reference
ISF, The Standard of Good Practice for Information Security 2016	IR2.6 Threat Profiling
National Institute of Standards and Technology Special Publication 800-53, Revision 5 (Draft), August 2017	3.7 Identification and authentication (IA-3); 3.11 Media protection (MP-1); 3.13 Physical and environmental protection (PE-5); 3.19 System and communications protection (SC-15)
The CIS Critical Security Controls for Effective Cyber Defense Version 6.1, August 2016	Maintenance, Monitoring, and Analysis of Audit Logs

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B. Component: Organizational Structures

Key Management Practice	Chief Information Officer	Chief Information Security Officer	Business Process Owners	Head Human Resources	Head Development	Head IT Operations	Information Security Manager	Privacy Officer
DSS05.01 Protect against malicious software.		A	R	R	R	R	R	
DSS05.02 Manage network and connectivity security.		A			R	R	R	
DSS05.03 Manage endpoint security.		A			R	R	R	
DSS05.04 Manage user identity and logical access.		A	R			R	R	R
DSS05.05 Manage physical access to I&T assets.		A				R	R	R
DSS05.06 Manage sensitive documents and output devices.	A					R		R
DSS05.07 Manage vulnerabilities and monitor the infrastructure for security-related events.		A				R	R	R

Related Guidance (Standards, Frameworks, Compliance Requirements)	Detailed Reference
No related guidance for this component	

C. Component: Information Flows and Items (see also Section 3.6)

Management Practice	Inputs		Outputs	
	From	Description	Description	To
DSS05.01 Protect against malicious software.			Malicious software prevention policy	APO01.02
			Evaluations of potential threats	APO12.02; APO12.03
DSS05.02 Manage network and connectivity security.	APO01.07	Data classification guidelines	Connectivity security policy	APO01.02
	APO09.03	SLAs	Results of penetration tests	MEA04.07
DSS05.03 Manage endpoint security.	APO03.02	Information architecture model	Security policies for endpoint devices	APO01.02
	APO09.03	• SLAs • OLAs		
	BAI09.01	Results of physical inventory checks		
	DSS06.06	Reports of violations		
DSS05.04 Manage user identity and logical access.	APO01.05	Definition of I&T-related roles and responsibilities	Results of reviews of user accounts and privileges	Internal
	APO03.02	Information architecture model	Approved user access rights	Internal

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C. Component: Information Flows and Items (see also Section 3.6) *(cont.)*				
Management Practice	**Inputs**		**Outputs**	
DSS05.05 Manage physical access to I&T assets.	**From**	**Description**	**Description**	**To**
			Access logs	DSS06.03, MEA04.07
			Approved access requests	Internal
DSS05.06 Manage sensitive documents and output devices.	APO03.02	Information architecture model	Access privileges	Internal
			Inventory of sensitive documents and devices	Internal
DSS05.07 Manage vulnerabilities and monitor the infrastructure for security-related events.			Security incident tickets	DSS02.02
			Security incident characteristics	Internal
			Security event logs	Internal
Related Guidance (Standards, Frameworks, Compliance Requirements)	**Detailed Reference**			
No related guidance for this component				

D. Component: People, Skills and Competencies		
Skill	**Related Guidance (Standards, Frameworks, Compliance Requirements)**	**Detailed Reference**
Information security	Skills Framework for the Information Age V6, 2015	SCTY
Information security management	e-Competence Framework (e-CF)—A common European Framework for ICT Professionals in all industry sectors—Part 1: Framework, 2016	E. Manage— E.8. Information Security Management
Penetration testing	Skills Framework for the Information Age V6, 2015	PENT
Security administration	Skills Framework for the Information Age V6, 2015	SCAD

E. Component: Policies and Procedures			
Relevant Policy	**Policy Description**	**Related Guidance**	**Detailed Reference**
Information security policy	Sets guidelines to protect corporate information and associated systems and infrastructure.		

F. Component: Culture, Ethics and Behavior		
Key Culture Elements	**Related Guidance**	**Detailed Reference**
Create a culture of awareness regarding user responsibility to maintain security and privacy practices.	1) HITRUST CSF version 9, September 2017; (2) ISF, The Standard of Good Practice for Information Security 2016	(1) 01.03 User Responsibilities; (2) PM2.1 Security Awareness Program

G. Component: Services, Infrastructure and Applications
• Directory services • Email filtering systems • Identity and access management system • Security awareness services • Security information and event management (SIEM) tools • Security operations center (SOC) services • Third-party security assessment services • URL filtering systems

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Domain: Deliver, Service and Support Management Objective: DSS06 - Managed Business Process Controls	Focus Area: COBIT Core Model

Description

Define and maintain appropriate business process controls to ensure that information related to and processed by in-house or outsourced business processes satisfies all relevant information control requirements. Identify the relevant information control requirements. Manage and operate adequate input, throughput and output controls (application controls) to ensure that information and information processing satisfy these requirements.

Purpose

Maintain information integrity and the security of information assets handled within business processes in the enterprise or its outsourced operation.

The management objective supports the achievement of a set of primary enterprise and alignment goals:

Enterprise Goals	Alignment Goals
• EG01 Portfolio of competitive products and services • EG05 Customer-oriented service culture • EG08 Optimization of internal business process functionality • EG12 Managed digital transformation programs	AG08 Enabling and supporting business processes by integrating applications and technology

Example Metrics for Enterprise Goals	Example Metrics for Alignment Goals
EG01 a. Percent of products and services that meet or exceed targets in revenues and/or market share b. Percent of products and services that meet or exceed customer satisfaction targets c. Percent of products and services that provide competitive advantage d. Time to market for new products and services	AG08 a. Time to execute business services or processes b. Number of I&T-enabled business programs delayed or incurring additional cost due to technology-integration issues c. Number of business process changes that need to be delayed or reworked because of technology-integration issues d. Number of applications or critical infrastructures operating in silos and not integrated
EG05 a. Number of customer service disruptions b. Percent of business stakeholders satisfied that customer service delivery meets agreed levels c. Number of customer complaints d. Trend of customer satisfaction survey results	
EG08 a. Satisfaction levels of board and executive management with business process capabilities b. Satisfaction levels of customers with service delivery capabilities c. Satisfaction levels of suppliers with supply chain capabilities	
EG12 a. Number of programs on time and within budget b. Percent of stakeholders satisfied with program delivery c. Percent of business transformation programs stopped d. Percent of business transformation programs with regular reported status updates	

A. Component: Process

Management Practice	Example Metrics
DSS06.01 Align control activities embedded in business processes with enterprise objectives. Continually assess and monitor the execution of business process activities and related controls (based on enterprise risk), to ensure that processing controls align with business needs.	a. Percent of completed inventory of critical processes and key controls b. Percent of processing controls aligned with business needs

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A. Component: Process *(cont.)*	
Activities	**Capability Level**
1. Identify and document the necessary control activities for key business processes to satisfy control requirements for strategic, operational, reporting and compliance objectives.	2
2. Prioritize control activities based on the inherent risk to the business. Identify key controls.	
3. Ensure ownership of key control activities.	
4. Implement automated controls.	3
5. Continually monitor control activities on an end-to-end basis to identify opportunities for improvement.	4
6. Continually improve the design and operation of business process controls.	5

Related Guidance (Standards, Frameworks, Compliance Requirements)	Detailed Reference
National Institute of Standards and Technology Special Publication 800-37, Revision 2 (Draft), May 2018	3.1 Preparation (Task 10, 11)
The CIS Critical Security Controls for Effective Cyber Defense Version 6.1, August 2016	CSC 14: Controlled Access Based on the Need to Know

Management Practice	Example Metrics
DSS06.02 Control the processing of information. Operate the execution of the business process activities and related controls, based on enterprise risk. Ensure that information processing is valid, complete, accurate, timely and secure (i.e., reflects legitimate and authorized business use).	a. Number of incidents and audit report findings indicating failure of key controls b. Percent of coverage of key controls within test plans

Activities	Capability Level
1. Authenticate the originator of transactions and verify that the individual has the authority to originate the transaction.	2
2. Ensure adequate segregation of duties regarding the origination and approval of transactions.	
3. Verify that transactions are accurate, complete and valid. Controls may include sequence, limit, range, validity, reasonableness, table look-ups, existence, key verification, check digit, completeness, duplicate and logical relationship checks, and time edits. Validation criteria and parameters should be subject to periodic reviews and confirmations. Validate input data and edit or, where applicable, send back for correction as close to the point of origination as possible.	3
4. Without compromising original transaction authorization levels, correct and resubmit data that were erroneously input. Where appropriate for reconstruction, retain original source documents for the appropriate amount of time.	
5. Maintain the integrity and validity of data throughout the processing cycle. Ensure that detection of erroneous transactions does not disrupt processing of valid transactions.	
6. Handle output in an authorized manner, deliver it to the appropriate recipient and protect the information during transmission. Verify the accuracy and completeness of the output.	
7. Maintain the integrity of data during unexpected interruptions in business processing. Confirm data integrity after processing failures.	
8. Before passing transaction data between internal applications and business/operational functions (inside or outside the enterprise), check for proper addressing, authenticity of origin and integrity of content. Maintain authenticity and integrity during transmission or transport.	

Related Guidance (Standards, Frameworks, Compliance Requirements)	Detailed Reference
HITRUST CSF version 9, September 2017	13.01 Openness and Transparency; 13.02 Individual Choice and Participation
ISF, The Standard of Good Practice for Information Security 2016	BA1.4 Information Validation

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A. Component: Process *(cont.)*	
Management Practice	**Example Metrics**
DSS06.03 Manage roles, responsibilities, access privileges and levels of authority. Manage business roles, responsibilities, levels of authority and segregation of duties needed to support the business process objectives. Authorize access to all information assets related to business information processes, including those under the custody of the business, IT and third parties. This ensures that the business knows where the data are and who is handling data on its behalf.	a. Number of incidents and audit findings due to access or separation-of-duties violations b. Percent of business process roles with assigned access rights and levels of authority c. Percent of business process roles with clear separation of duties

Activities	Capability Level
1. Allocate roles and responsibilities based on approved job descriptions and business process activities.	2
2. Allocate levels of authority for approval of transactions, transaction limits and any other decisions relating to the business process, based on approved job roles.	
3. Allocate roles for sensitive activities so there is a clear segregation of duties.	
4. Allocate access rights and privileges based on the minimum that is required to perform job activities, based on pre-defined job roles. Remove or revise access rights immediately if the job role changes or a staff member leaves the business process area. Periodically review to ensure that the access is appropriate for the current threats, risk, technology and business need.	3
5. On a regular basis, provide awareness and training regarding roles and responsibilities so that everyone understands their responsibilities; the importance of controls; and the security, integrity, confidentiality and privacy of company information in all its forms.	
6. Ensure administrative privileges are sufficiently and effectively secured, tracked and controlled to prevent misuse.	
7. Periodically review access control definitions, logs and exception reports. Ensure that all access privileges are valid and aligned with current staff members and their allocated roles.	4

Related Guidance (Standards, Frameworks, Compliance Requirements)	Detailed Reference
HITRUST CSF version 9, September 2017	13.04 Collection, Use and Disclosure
ISO/IEC 27002:2013/Cor.2:2015(E)	7. Human resource security
The CIS Critical Security Controls for Effective Cyber Defense Version 6.1, August 2016	CSC 5: Controlled Use of Administrative Privileges

Management Practice	Example Metrics
DSS06.04 Manage errors and exceptions. Manage business process exceptions and errors and facilitate remediation, executing defined corrective actions and escalating as necessary. This treatment of exceptions and errors provides assurance of the accuracy and integrity of the business information process.	a. Frequency of processing inefficiencies due to incomplete data entry b. Number of errors detected in a timely manner c. Number of data processing errors that were efficiently remediated

Activities	Capability Level
1. Review errors, exceptions and deviations.	2
2. Follow up, correct, approve and resubmit source documents and transactions.	
3. Maintain evidence of remedial actions.	
4. Define and maintain procedures to assign ownership for errors and exceptions, correct errors, override errors and handle out-of-balance conditions.	3
5. Report relevant business information process errors in a timely manner to perform root cause and trending analysis.	4

Related Guidance (Standards, Frameworks, Compliance Requirements)	Detailed Reference
No related guidance for this management practice	

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A. Component: Process *(cont.)*		
Management Practice	**Example Metrics**	
DSS06.05 Ensure traceability and accountability for information events. Ensure that business information can be traced to an originating business event and associated with accountable parties. This discoverability provides assurance that business information is reliable and has been processed in accordance with defined objectives.	a. Number of incidents in which transaction history cannot be recovered b. Percent of completeness of traceable transaction log	
Activities		**Capability Level**
1. Capture source information, supporting evidence and the record of transactions.		2
2. Define retention requirements, based on business requirements, to meet operational, financial reporting and compliance needs.		3
3. Dispose of source information, supporting evidence and the record of transactions in accordance with the retention policy.		
Related Guidance (Standards, Frameworks, Compliance Requirements)	**Detailed Reference**	
No related guidance for this management practice		
Management Practice	**Example Metrics**	
DSS06.06 Secure information assets. Secure information assets accessible by the business through approved methods, including information in electronic form (e.g., portable media devices, user applications and storage devices, or other methods that create new assets in any form), information in physical form (e.g., source documents or output reports) and information during transit. This benefits the business by providing end-to-end safeguarding of information.	a. Cases of sensitive transaction data delivered to wrong recipient b. Frequency of compromised integrity of critical data	
Activities		**Capability Level**
1. Restrict use, distribution and physical access of information according to its classification.		2
2. Provide acceptable use awareness and training.		
3. Apply data classification and acceptable use and security policies and procedures to protect information assets under the control of the business.		3
4. Identify and implement processes, tools and techniques to reasonably verify compliance.		
5. Report to business and other stakeholders on violations and deviations.		4
Related Guidance (Standards, Frameworks, Compliance Requirements)	**Detailed Reference**	
CMMI Cybermaturity Platform, 2018	AC.MP Manage Access Permissions	
The CIS Critical Security Controls for Effective Cyber Defense Version 6.1, August 2016	CSC 18: Application Software Security	

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B. Component: Organizational Structures

Key Management Practice	Executive Committee	Chief Information Officer	I&T Governance Board	Chief Information Security Officer	Business Process Owners	Data Management Function	Service Manager	Information Security Manager	Legal Counsel
DSS06.01 Align control activities embedded in business processes with enterprise objectives.	R		A		R				
DSS06.02 Control the processing of information.		R	A	R	R	R			R
DSS06.03 Manage roles, responsibilities, access privileges and levels of authority.		R	A	R	R			R	
DSS06.04 Manage errors and exceptions.		R		R	A		R		
DSS06.05 Ensure traceability and accountability for information events.		R		R	A				
DSS06.06 Secure information assets.		R		R	A				

Related Guidance (Standards, Frameworks, Compliance Requirements)	Detailed Reference
No related guidance for this component	

C. Component: Information Flows and Items (see also Section 3.6)

Management Practice	Inputs		Outputs	
	From	Description	Description	To
DSS06.01 Align control activities embedded in business processes with enterprise objectives.	APO01.07	• Data classification guidelines • Data integrity procedures	Root cause analyses and recommendations	BAI06.01; MEA02.04; MEA04.04; MEA04.06; MEA04.07
			Results of processing effectiveness reviews	MEA02.04
DSS06.02 Control the processing of information.	BAI05.05	Operation and use plan	Processing control reports	Internal
	BAI07.02	Migration plan		
DSS06.03 Manage roles, responsibilities, access privileges and levels of authority.	APO11.01	Quality management system (QMS) roles, responsibilities and decision rights	Allocated levels of authority	APO01.05
	APO13.01	Information security management system (ISMS) scope statement	Allocated roles and responsibilities	APO01.05
	DSS05.05	Access logs	Allocated access rights	APO07.04
	EDM04.02	Assigned responsibilities for resource management		

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C. Component: Information Flows and Items (see also Section 3.6) *(cont.)*				
Management Practice	**Inputs**		**Outputs**	
DSS06.04 Manage errors and exceptions.	**From**	**Description**	**Description**	**To**
			Error reports and root cause analysis	Internal
			Evidence of error correction and remediation	MEA02.04
DSS06.05 Ensure traceability and accountability for information events.			Record of transactions	Internal
			Retention requirements	Internal; APO14.09
DSS06.06 Secure information assets.			Reports of violations	DSS05.03

Related Guidance (Standards, Frameworks, Compliance Requirements)	Detailed Reference
National Institute of Standards and Technology Special Publication 800-37, Revision 2, September 2017	3.1 Preparation (Task 10, 11): Inputs and Outputs

D. Component: People, Skills and Competencies		
Skill	**Related Guidance (Standards, Frameworks, Compliance Requirements)**	**Detailed Reference**
Information security	Skills Framework for the Information Age V6, 2015	SCTY
Security administration	Skills Framework for the Information Age V6, 2015	SCAD

E. Component: Policies and Procedures			
Relevant Policy	**Policy Description**	**Related Guidance**	**Detailed Reference**
Business controls guidance	Defines business process controls to ensure proper control and reduce risk of fraud and errors. Identifies manual controls to protect documents (e.g., source, input, processing and output documents); identifies supervisory controls to review the flow of documents and ensure correct processing. Includes I&T general controls (e.g., physical security, access and authentication, and change management) and application controls (e.g., edit checking, system configuration and security settings).		

F. Component: Culture, Ethics and Behavior		
Key Culture Elements	**Related Guidance**	**Detailed Reference**
Create a culture that embraces the need for sound controls in business processes, building them into applications in development or requiring them in applications bought or accessed as a service. Encourage all employees to have a controls consciousness to protect all assets of the organization (e.g., paper records and facilities).		

G. Component: Services, Infrastructure and Applications
• Automated application controls • Event log auditing tools

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